

UNIVERSITI TEKNOLOGI MALAYSIA

PROJECT 1 (15%)

SEMESTER II 2021/2022

CODE OF SUBJECT : SCSJ4383/ SCJ4383
NAME OF SUBJECT : SOFTWARE CONSTRUCTION
YEAR / COURSE : 3/4/SCJ/4SCJ

ASSIGNMENTS:

- **PART A:** Application of DevOps
- **PART B:** Design of code, design smells, and Refactoring

INSTRUCTION:

1. The submission deadline: Week 15

PART A: DevOps (10%)

1. Create a Project Account in Jira PM Cloud and invites the team members and Instructor/s as a collaborator in the Project Account
2. Create a Project Repo in Github and invites the team members and Instructor/s as a collaborator in the Project Repo
3. Integrate Jira, Github, Jmeter, and Docker Plugin (Install and setup) to your Jenkins.
4. Create an issue in Jira Project Account and update the issue in your Jenkins Project Pipeline.
5. Commit and push your project to Github repo, build run/ update in your Jenkins Project Pipeline.
6. All the Performance Testing reports from JMeter will be updated in your Jenkins Project Pipeline.
7. Create Docker Image from the Github Project repo using Jenkins and push it to Docker Hub.
8. Other Team members pull the Docker Image from the Docker Hub to his /her workstation using Docker.
9. All the steps above must be recorded and submitted to e-learning by the stated due date

PART B: Code Smells and Refactoring (5%)

1. Find OR develop and present existing software **codes** from any sources such as Final year past projects that you can reach out.
2. List out at least **TWO (2) code smell** from the existing software.
3. Explains the methods of detecting code smells based on Question no. 2.
4. Explains the methods of code re-factoring.
5. Fix and re-factored (based on Question no. 1 (PART B)) and re-run the system again after re-factoring.
6. BONUS MARKS. Compare the performance (execution time) of the original code and refactored code using the available tools. (This is an option, and your group will get extra marks if successfully completed this task)

SUBMISSION:

Part A:

1. **Presentation Slide** for items 1-8.
2. **Video Presentation** for items 1-8.
3. Source code: **Github Link**

Part B:

1. **Documentation** (pdf)
2. Source code: **Github Link**

DUE DATE:

1. **Part A:** Week 15 (Please refer to E-Learning for a specific date)
2. **Part B:** Final Examination Week (Please refer to E-Learning for specific date)